

Echara — Product & Technical Specification (v1.1, MVP)

Admin: Web-only

AI recommendations: Reserved for later phase

Platforms: Cross-platform mobile (iOS/Android) + Web (buyer & vendor) + Web Admin

Region: UK only

0) Branding & Design Guardrails

- **Name:** Echara
- **Logo (placeholder):** Wedding cake icon
- **Palette:** Cream #f4f3ea (70%), Cognac #9c7b70 (20%), Honeysuckle #d03952 (10%)
- **Buyer UX:** Mirror Airbnb flow and patterns with our palette.
- **Vendor UX:** Mirror Monzo dashboard patterns (cards, insights, rounded visualisations).
- **Typography:** Inter / SF Pro Display / Poppins
- **A11y:** WCAG AA; text contrast $\geq 4.5:1$; keyboard nav; labels for form inputs.

1) Architecture Overview

1.1 Stack (MVP, free-tier friendly)

- **Mobile:** React Native (Expo)
- **Web:** Next.js (App Router)
- **Backend:** Node.js + Express (hosted on Vercel serverless functions where feasible; long-lived sockets on a minimal Node instance, e.g., a single small VM/Render/Dokku)
- **Realtime:** Socket.IO (separate small Node process)
- **DB:** MongoDB Atlas (EU region)
- **Auth:** Firebase Auth (Email/Password, Google, Apple)
- **Storage:** Firebase Storage (images, docs)
- **Push:** Firebase Cloud Messaging (mobile + web push)
- **Email:** Resend or SendGrid (free tier)

- **Maps/Geo:** Google Maps Platform (Places + Maps + Geocoding)
- **Calendar sync:** Google Calendar API (OAuth)
- **Admin:** Next.js web app behind role-gated access
- **CI/CD:** Vercel (web + APIs) + EAS (Expo) for mobile builds

Note: For Socket.IO (stateful), host a tiny Node process separate from Vercel. Keep messages also persisted in MongoDB.

1.2 Environments

- **dev** (shared sandbox)
- **stage** (pre-prod, limited data)
- **prod** (real users)

1.3 Env Vars (indicative)

```
FIREBASE_PROJECT_ID=
FIREBASE_CLIENT_EMAIL=
FIREBASE_PRIVATE_KEY=
NEXT_PUBLIC_FIREBASE_API_KEY=
MONGODB_URI=
JWT_SECRET=
GCLOUD_MAPS_API_KEY=
RESEND_API_KEY= # or SENDGRID_API_KEY
FCM_SERVER_KEY=
GOOGLE_OAUTH_CLIENT_ID=
GOOGLE_OAUTH_CLIENT_SECRET=
APP_BASE_URL=
```

2) Roles & Permissions

- **Buyer:** browse, filter, wishlist, request bookings, chat, review after completion.
- **Vendor:** create/manage business profile, create listings, approve/decline bookings, chat, view insights, manage subscription (trial → £29/month), connect Google Calendar, receive email alerts.
- **Admin (web-only):** approve/reject vendors, manage listings, moderate reviews & photos, view platform analytics, access dispute view (read-only chat), suspend accounts, toggle “featured”.

AuthZ model: Role claims issued post-sign-in (Firebase custom claims or server-side role lookup). All APIs enforce role checks.

3) Functional Modules & Acceptance Criteria

3.1 Authentication & Profile

- **Sign-in methods:** Email/Password, Google, Apple.
- **AC:**
 - Able to register and sign in/out on all platforms.
 - Role assigned at first vendor onboarding.
 - Session refresh without losing state.
 - GDPR: endpoint to export and delete user data.

3.2 Vendor Onboarding & Business Profile

Fields: businessName, yearsInBusiness, description, tags[], socialLinks{instagram, tiktok, website}, address + geocode, heroImage, gallery[], trialEndDate, subscriptionStatus(trial|active|past_due|canceled).

- **AC:**
 - Vendor can complete onboarding wizard and save a draft.
 - Trial starts at first activation; free for 60 days; hard-block paid features post-trial (but allow profile edits).
 - Admin can mark vendor as “approved” before listings go public (soft trust even though no document verification).

3.3 Listings

Fields: vendorId, category, priceRange{min,max,currency}, description, gallery[], location{lat,lng,address}, availability[], policies{depositPercent, cancellationWindow, refundRules}, ratingSummary{avgOverall, count}.

- **AC:**
 - Create/edit/delete listing, upload media, validate categories.
 - Listing visible in search upon Admin approval (configurable: “auto-publish on vendor approval”).
 - Map pin displays correctly; directions link opens map.

- Availability preview reflects connected calendar (read-only import in MVP).

3.4 Search & Discovery (Buyer)

- **Filters:** category, price range, location radius, rating, availability window.
- **Sort:** relevance, rating, price, most reviewed.
- **AC:**
 - Query returns paginated results < 300ms p95 on stage data.
 - Empty-state and no-result states implemented.
 - “Save to Wishlist” persists per user.

3.5 Booking Requests (No in-app payment)

Status flow: pending -> confirmed | declined -> completed | canceled

- **AC:**
 - Buyer can propose date/time from vendor slots (pre-filled from vendor availability or connected calendar).
 - Vendor receives push + email; can approve/decline.
 - Booking timeline shows state changes with timestamps.
 - Payment fields are **logging only**: vendor can mark amount/method/date for record; no processing in app.
 - Cancellations respected with reason capture (buyer/vendor sides).
 - Completion can be auto-prompted by date + vendor confirmation.

3.6 Messaging (Realtime)

- **Features:** socket rooms per buyer↔vendor pair, message types text|image|file, typing, read receipts.
- **AC:**
 - Messages persist in MongoDB; offline delivery works.
 - Rate-limit media size (e.g., 10MB) and file types (jpg/png/pdf).
 - Chat is available only after a buyer initiates an enquiry or booking request.

3.7 Reviews & Ratings

- **Who:** Buyer after “completed” booking.
- **Criteria:** quality, communication, value, professionalism (1–5 stars), optional text.

- **Vendor reply:** single public response, editable within 24 hours.
- **AC:**
 - One review per booking.
 - Aggregate ratings recalculated on write.
 - Admin can hide reviews violating policy (retain audit log).

3.8 Vendor Insights (Monzo-like)

- **KPIs:** earnings overview (logged), bookings & enquiries, cancellation rate, conversion rate (views→bookings), ad spend (manual entry in MVP), top listings, basic buyer geo/demographics (from profiles).
- **AC:**
 - Month/quarter toggles; charts render in < 500ms once data loaded.
 - All metrics backed by aggregation pipelines with test fixtures.

3.9 Notifications

- **Push:** booking status changes, new messages, review posted, trial ending, subscription status changes.
- **Email (vendors):** booking requests, daily digest, trial reminders, subscription receipts (when Stripe added).
- **AC:**
 - Deduplicated notifications; in-app bell shows unread counts.
 - User can mute categories.

3.10 Admin (Web-only)

- **Vendor mgmt:** approve/reject, suspend, feature.
- **Listing mgmt:** edit/hide/remove media.
- **Moderation:** reviews/photos flag queue.
- **Analytics:** active vendors/buyers, bookings, category heatmaps, rating distribution, growth curves.
- **Disputes:** read-only chat view, manual notes.
- **AC:**
 - All admin actions logged in AdminLogs with who/when/what.
 - Role-based access; admin portal not publicly indexed.

3.11 Subscription (Vendor)

- **Plan:** £29/month after 60-day free trial.

- **MVP billing:** optionally stubbed. If Stripe not integrated, show “Upgrade” flow as mock with manual flag by admin.
- **AC:**
 - Trial countdown visible; post-trial gates insights & booking approvals until plan is active (configurable which features lock).

3.12 Phase 2 Placeholder (Not in MVP)

- **AI Recommendations:** “Vendors you may like” using category, location, behaviour; embeddings later.
- **Stripe Connect:** escrow or marketplace payouts.
- **CSV exports for tax.**
- **Sponsored listings/ads.**

4) Data Model (Mongo, high-level)

// users

```
{
  _id, role: 'buyer'|'vendor'|'admin',
  name, email, avatarUrl, favourites: [listingId],
  createdAt, updatedAt, gdprDeletedAt?
}
```

// vendors

```
{
  _id, userId, businessName, yearsInBusiness, description,
  tags: [string], socialLinks: { instagram?, tiktok?, website? },
  address, geo: { lat, lng }, gallery: [url], heroImage,
  status: 'pending'|'approved'|'suspended',
  subscriptionStatus: 'trial'|'active'|'past_due'|'canceled',
  trialEndDate, adSpendMonthly?: number,
  createdAt, updatedAt
}
```

// listings

```
{
  _id, vendorId, category, priceRange: { min, max, currency: 'GBP' },
  description, gallery: [url],
  location: { lat, lng, address },
}
```

```
availability: [{ startISO, endISO, source: 'manual'|'google' }],
policies: { depositPercent?, cancellationWindowDays?, refundRules? },
ratingSummary: { avgOverall, count, criteriaAvg: { quality, communication, value,
professionalism } },
isFeatured: boolean, isVisible: boolean,
createdAt, updatedAt
}
```

```
// bookings
```

```
{
  _id, buyerId, vendorId, listingId,
  status: 'pending'|'confirmed'|'declined'|'completed'|'canceled',
  requestedSlot: { startISO, endISO },
  approvedAt?, declinedAt?, completedAt?, canceledAt?, cancelReason?,
  paymentLogged?: { amountGBP, method: 'cash'|'bank'|'other', dateISO, notes? },
  messagesThreadId,
  createdAt, updatedAt
}
```

```
// chats
```

```
{
  _id, buyerId, vendorId, listingId,
  messages: [{ _id, senderId, type: 'text'|'image'|'file', contentUrlOrText, createdAt,
seenBy: [userId] }],
  lastMessageAt
}
```

```
// reviews
```

```
{
  _id, bookingId, buyerId, vendorId, listingId,
  ratings: { quality, communication, value, professionalism },
  overall: number, comment, vendorReply?, repliedAt?,
  createdAt, updatedAt, hiddenByAdmin?: boolean
}
```

```
// notifications
```

```
{
  _id, userId, type, title, body, data, isRead, createdAt
}
```

```
// admin logs
```

```
{
  _id, performedByAdminId, action, targetType, targetId, metadata, createdAt
}
```

Indexes (examples):

- listings: { category: 1, "location.lat": 1, "location.lng": 1 }, text index on description, tags.
- bookings: { vendorId: 1, status: 1, requestedSlot.startISO: 1 }.
- chats: { buyerId: 1, vendorId: 1, lastMessageAt: -1 }.
- reviews: { vendorId: 1, createdAt: -1 }.

5) Public API (REST) — Outline

Base path: /api/v1

Auth & Profile

- POST /auth/register (email/pw)
- POST /auth/login
- POST /auth/oauth/callback (Google/Apple)
- GET /me — current user & role
- DELETE /me — GDPR delete
- GET /me/favourites / POST /me/favourites/:listingId

Vendors

- POST /vendors — create/update profile (role-guarded)
- GET /vendors/:id
- GET /vendors/:id/insights?range=month|quarter
- POST /vendors/:id/calendar/google/connect
- DELETE /vendors/:id/calendar/google/disconnect

Listings

- POST /listings (vendor)
- PUT /listings/:id (vendor)
- DELETE /listings/:id (vendor/admin)
- GET /listings/:id

- GET
/listings/search?category&lat&lng&radius&minPrice&maxPrice&rating&availableFrom&availableTo&sort&page&limit

Bookings

- POST /bookings — buyer creates request
- GET /bookings/:id
- POST /bookings/:id/approve (vendor)
- POST /bookings/:id/decline (vendor)
- POST /bookings/:id/cancel (buyer/vendor)
- POST /bookings/:id/complete (vendor)
- POST /bookings/:id/payment-log (vendor)

Chat (REST + Sockets)

- GET /chats?as=buyer|vendor — list conversations
- GET /chats/:id — history (paginated)
- **Socket events:**
 - chat:join {chatId}
 - chat:message {chatId, type, payload}
 - chat:typing {chatId}
 - chat:seen {chatId, messageId}

Reviews

- POST /reviews — after completed booking
- GET /vendors/:id/reviews?page&limit
- POST /reviews/:id/reply (vendor)
- DELETE /reviews/:id (admin hide → sets hiddenByAdmin)

Notifications

- GET /notifications
- POST /notifications/:id/read

Admin (web-only; role-guarded)

- GET /admin/vendors?status=pending|approved|suspended
- POST /admin/vendors/:id/status
- POST /admin/listings/:id/visibility
- POST /admin/reviews/:id/hide
- GET /admin/analytics/overview?from&to

- GET /admin/logs?page&limit

6) Validation & Business Rules (selected)

- **Vendor approval** required before any listings are publicly searchable.
- **Booking conflict check:** vendor cannot approve overlapping confirmed bookings.
- **Review eligibility:** only for completed bookings, 30-day window to review.
- **Media upload:** virus/mime validation server-side; max 10MB per file.
- **Rate limits:** chat messages (e.g., 30/min per user), search (e.g., 60/min/IP).
- **GDPR delete:** anonymise PII; retain booking aggregations with detached IDs.

7) Notifications Matrix (sample)

Trigger	Recipient	Push	Email	In-App
Booking requested	Vendor	✓	✓	✓
Booking approved/declined	Buyer	✓	—	✓
New chat message	Both	✓	(vendor digest optional)	✓
Review posted	Vendor	✓	✓	✓
Trial ending in 7 days	Vendor	✓	✓	✓
Account suspended (admin)	Vendor	✓	✓	✓

8) Vendor Insights — Aggregations

- **Earnings overview:** sum(bookings.paymentLogged.amountGBP) by month/quarter.
- **Bookings & enquiries:** count bookings by status, time.
- **Cancellation rate:** canceled / confirmed.
- **Conversion rate:** (confirmed bookings) / (listing views). Track ListingViews as a light collection { listingId, userId?, ts }.
- **Ad spend:** manual input retained per month.
- **Top listings:** by confirmed bookings or page views.
- **Buyer geo:** from buyer profile or IP-to-geo (rough; optional).

9) Security & Compliance

- **Transport:** HTTPS only; HSTS.
- **Auth:** Firebase ID tokens verified on server; JWT for server-issued sessions if needed.
- **Input sanitisation:** zod/yup on all request bodies.
- **Upload safety:** mime sniff + size limits + signed URLs.
- **Secrets:** .env, never in repo.
- **GDPR:** privacy policy, DSR (export/delete), data minimisation, EU data region on Atlas.
- **Abuse:** content reporting, IP throttling, photo moderation toggle (admin hide).

10) Performance, Observability, Testing

- **API p95:** < 300ms for search/read ops on stage data.
- **Caching:** CDN cache for listing pages & images; short-TTL search cache (e.g., 30s mem cache).
- **Logs:** structured JSON logs; error tracking via Sentry (free tier).
- **Tests:**
 - Unit tests for services & validators
 - Integration tests for core flows (auth, listing CRUD, booking flow, chat)
 - E2E smoke (Playwright) for web, basic Detox for mobile critical paths

11) Deployment & Release

- **Web (buyer/vendor/admin):** Vercel preview → protected stage → prod.
- **API:** Vercel functions (stateless) + tiny Node host for Sockets.
- **Mobile:** Expo EAS builds; internal testing via TestFlight & Play Console.
- **Feature flags:** simple collection FeatureFlags to toggle admin-only features.

12) Category Set (seed for UK weddings & events)

Example categories (extensible):

Venues, Photographers, Videographers, Caterers, Cake Makers, Florists, DJs, Live Bands, Decor & Styling, Car Hire, Hair & Makeup, Suits, Dresses, Invitations, Lighting & AV, Marquees & Tents, Event Planners, Officiants, Entertainment (magicians etc.), Photo Booths, Favours & Gifts, Security, Clean-up

You can prune further later. The user-facing filters should group smartly (e.g., “Food & Drink”, “Music & Entertainment”, “Beauty & Attire”, “Venues & Spaces”, “Planning & Styling”).

13) Admin Analytics (Web-only)

Dashboards:

- **Overview:** Active buyers/vendors, new signups, bookings requested/confirmed, cancellations, top categories.
- **Geography:** heatmap of listings and bookings by UK region.
- **Quality:** average ratings by category; flag vendors < 3.2 stars or with high cancel rate.
- **Growth:** MAU/WAU/DAU, conversion funnel (view → request → confirm).
- **Ops:** moderation queue length, SLA to decision, dispute notes.

14) MVP Cutline (what's IN vs OUT)

IN:

- Cross-platform apps + web
- Real-time chat & media
- Booking approvals (no payments)
- Reviews & vendor replies
- Vendor insights (without CSV export)
- Push + vendor email notifications
- Google Calendar read-only sync
- Admin web-only portal

OUT (Phase 2+):

- AI recommendations
- Stripe/escrow payments
- CSV exports for tax
- Sponsored listings/ads
- Sophisticated fraud detection

15) Open Questions / Config Decisions

- **Auto-publish policy:** After vendor approval, should new listings auto-publish or require per-listing approval? (*Default: auto-publish; admin can hide.*)
- **Availability source of truth:** If Google Calendar connected, do we **block** manual slots that conflict or just flag? (*Default: block.*)
- **Post-trial gating:** Which features are blocked when trial ends and subscription inactive? (*Default: booking approval & insights locked; profile/listing edit allowed.*)